

Processando config 23, seed 2, teste: Rabbit  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
Starting Rabbit: nb = 16777216  
Version: TestU01 1.2.3  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

smultin\_MultinomialBitsOver test:

-----  
N = 2, n = 8388608, r = 0, s = 32, L = 38,  
Sparse = TRUE

Number of bits = n = 8388608  
Number of cells =  $2^L$  = 274877906944  
Expected number per cell =  $1 / 32768$   
EColl =  $n^2 / (2k)$  = 128  
Hashing = TRUE

Collision test  
CollisionOver: density =  $n / k = 1 / 32768.00$   
Expected number of collisions =  $\mu = 128.00$

-----  
Results of CollisionOver test:

POISSON approximation :  
Expected number of collisions =  $N \cdot \mu$  : 256.00  
Observed number of collisions : 261  
p-value of test : 0.39

-----  
Total number of cells containing j balls

|       |   |              |
|-------|---|--------------|
| j = 0 | : | 549739036933 |
| j = 1 | : | 16776694     |
| j = 2 | : | 261          |
| j = 3 | : | 0            |
| j = 4 | : | 0            |
| j = 5 | : | 0            |

-----  
CPU time used : 00:00:04.49

Generator state:  
16777216 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 262144, r = 0, t = 2

-----  
Max of all bit distances : 17.00  
p-value of test : 0.39

-----  
CPU time used : 00:00:00.04

Generator state:  
33554432 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

snpair\_ClosePairsBitMatch test:

-----  
N = 1, n = 131072, r = 0, t = 4

-----  
Max of all bit distances : 8.00  
p-value of test : 0.12

-----  
CPU time used : 00:00:00.08

Generator state:  
50331648 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

svaria\_AppearanceSpacings test:

-----  
N = 1, Q = 524288, K = 524288, r = 0, s = 30, L = 15

Sequences of  $n = (K + Q)L = 15728640$  bits  
Q = 524288 initialization blocks  
K = 524288 blocks for the test  
the blocks have L = 15 bits

-----  
Normal statistic : -0.17  
p-value of test : 0.57

-----  
CPU time used : 00:00:00.00

Generator state:  
67108864 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

scomp\_LinearComp test:

-----  
N = 1, n = 18176, r = 0, s = 32

-----  
Number of degrees of freedom : 7  
Chi2 statistic for size of jumps : 8.72  
p-value of test : 0.27

-----  
Normal statistic for number of jumps : -51.05  
p-value of test : 1 - eps1 \*\*\*\*\*

-----  
CPU time used : 00:00:00.05

Generator state:  
67127040 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

scomp\_LempelZiv test:

-----  
N = 1, n = 16777216, r = 0, s = 32, k = 24

-----  
Normal statistic : -1.31  
p-value of test : 0.90

-----  
CPU time used : 00:00:00.41

Generator state:

83904256 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sspectral\_Fourier1 test:

-----

N = 1, n = 1048576, r = 0, s = 32, k = 20

-----

Normal statistic : 1.26  
p-value of test : 0.10

-----

CPU time used : 00:00:00.20

Generator state:

84952832 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sspectral\_Fourier3 test:

-----

N = 8065, n = 2048, r = 0, s = 32, k = 11

-----

Kolmogorov-Smirnov+ statistic = D+ : 0.038  
p-value of test : 0.22

Kolmogorov-Smirnov- statistic = D- : 0.010  
p-value of test : 0.90

Anderson-Darling statistic = A2 : 1.05  
p-value of test : 0.33

-----

CPU time used : 00:00:00.12

Generator state:

101728032 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_LongestHeadRun test:

-----  
N = 1, n = 60, r = 0, s = 32, L = 279616

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 3.52  
p-value of test : 0.32

-----  
Global longest run of 1 : 23.00  
p-value of test : 0.50

-----  
CPU time used : 00:00:00.05

Generator state:  
118504992 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_PeriodsInStrings test:

-----  
N = 1, n = 524288, r = 0, s = 31

-----  
Number of degrees of freedom : 94  
Chi-square statistic : 82.11  
p-value of test : 0.80

-----  
CPU time used : 00:00:00.02

Generator state:  
135282208 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_HammingWeight test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Number of degrees of freedom : 22  
Chi-square statistic : 15.69  
p-value of test : 0.83  
-----

-----  
CPU time used : 00:00:00.00

Generator state:  
152059424 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 524288, r = 0, s = 32, L = 32

-----  
Normal statistic : 0.55  
p-value of test : 0.29  
-----

-----  
CPU time used : 00:00:00.00

Generator state:  
168836640 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_HammingCorr test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 64

-----  
Normal statistic : -2.72  
p-value of test : 0.9967  
-----

-----  
CPU time used : 00:00:00.00

Generator state:  
185613856 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_HammingCorr test:

-----

N = 1, n = 131072, r = 0, s = 32, L = 128

-----

Normal statistic : -2.36  
p-value of test : 0.9909

-----

CPU time used : 00:00:00.00

Generator state:  
202391072 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_HammingIndep test:

-----

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

-----

Number of degrees of freedom : 177  
Chi-square statistic : 167.52  
p-value of test : 0.68

-----

CPU time used : 00:00:00.01

Generator state:  
219168288 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 305  
Chi-square statistic : 301.03  
p-value of test : 0.55  
-----

CPU time used : 00:00:00.00

Generator state:

235945504 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_HammingIndep test:

-----  
N = 1, n = 131072, r = 0, s = 32, L = 64, d = 0

Counters with expected numbers >= 10

-----  
Number of degrees of freedom : 489  
Chi-square statistic : 540.63  
p-value of test : 0.05  
-----

CPU time used : 00:00:00.00

Generator state:

252722720 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

sstring\_AutoCor test:

-----  
N = 1, n = 16777153, r = 0, s = 32, d = 1

```
-----
Normal statistic          :   -0.27
p-value of test          :    0.61
-----
```

```
CPU time used            :   00:00:00.01
```

```
Generator state:
 269499904 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
```

```
sstring_AutoCor test:
```

```
-----
      N =  1,   n = 16777154,   r =  0,   s = 32,   d = 2
-----
```

```
-----
Normal statistic          :    1.18
p-value of test          :    0.12
-----
```

```
CPU time used            :   00:00:00.00
```

```
Generator state:
 286277088 bits have been read.
```

```
*****
```

```
HOST = matheus-IdeaPad-3-15ALC6, Linux
```

```
ufile_CreateReadBin:   /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config23_seed2.bin
```

```
sstring_Run test:
```

```
-----
      N =  1,   n = 3355443,   r =  0,   s =  32
-----
```

```
-----
Total number of 1 runs:  3355449
```

```
Number of degrees of freedom      :   36
Chi2 statistic for number of runs :  42.45
p-value of test                   :    0.21
-----
```

```
-----
Total number of bits:  13422240
```

```
Normal statistic for number of bits :   0.090
p-value of test                   :    0.46
```

-----  
CPU time used : 00:00:00.04

Generator state:  
299699328 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 16384, r = 0, s = 32, L = 32, k = 32

-----  
Number of degrees of freedom : 3  
Chi-square statistic : 3.60  
p-value of test : 0.31

-----  
CPU time used : 00:00:00.08

Generator state:  
316476544 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

smarsa\_MatrixRank test:

-----  
N = 1, n = 163, r = 0, s = 32, L = 320, k = 320

-----  
Number of degrees of freedom : 2  
Chi-square statistic : 1056.73  
p-value of test : eps \*\*\*\*\*

-----  
CPU time used : 00:00:00.03

Generator state:  
333167744 bits have been read.

\*\*\*\*\*

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 131072, r = 0, s = 32, L0 = 128, L1 = 128

-----  
Test on the values of the Statistic H

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 42    |
| ChiSquare statistic          | : | 34.00 |
| p-value of test              | : | 0.81  |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 42    |
| ChiSquare statistic          | : | 30.08 |
| p-value of test              | : | 0.92  |

-----  
Test on the values of the Statistic J

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 64    |
| ChiSquare statistic          | : | 52.34 |
| p-value of test              | : | 0.85  |

-----  
Test on the values of the Statistic R

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 36    |
| ChiSquare statistic          | : | 22.99 |
| p-value of test              | : | 0.95  |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 22    |
| ChiSquare statistic          | : | 17.32 |
| p-value of test              | : | 0.75  |

-----  
CPU time used : 00:00:00.04

Generator state:

349944960 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 16384, r = 0, s = 32, L0 = 1024, L1 = 1024

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 94     |
| ChiSquare statistic          | : | 121.79 |
| p-value of test              | : | 0.03   |

-----  
Test on the values of the Statistic M

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 94    |
| ChiSquare statistic          | : | 91.23 |
| p-value of test              | : | 0.56  |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 512    |
| ChiSquare statistic          | : | 473.09 |
| p-value of test              | : | 0.89   |

-----  
Test on the values of the Statistic R

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 90     |
| ChiSquare statistic          | : | 102.09 |
| p-value of test              | : | 0.18   |

-----  
Test on the values of the Statistic C

|                              |   |       |
|------------------------------|---|-------|
| Number of degrees of freedom | : | 50    |
| ChiSquare statistic          | : | 69.36 |
| p-value of test              | : | 0.04  |

-----  
CPU time used : 00:00:00.03

Generator state:

366722176 bits have been read.

\*\*\*\*\*  
HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile\_CreateReadBin: /home/matheus/Área de  
Trabalho/qanalog\_prng-main/qanalog\_prng-main/outputs/out\_config23\_seed2.bin

swalk\_RandomWalk1 test:

-----  
N = 1, n = 1675, r = 0, s = 32, L0 = 10016, L1 = 10016

-----  
Test on the values of the Statistic H

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 134.49 |
| p-value of test              | : | 0.38   |

-----  
Test on the values of the Statistic M

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 131    |
| ChiSquare statistic          | : | 136.54 |
| p-value of test              | : | 0.35   |

-----  
Test on the values of the Statistic J

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 161    |
| ChiSquare statistic          | : | 166.14 |
| p-value of test              | : | 0.37   |

-----  
Test on the values of the Statistic R

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 130    |
| ChiSquare statistic          | : | 118.59 |
| p-value of test              | : | 0.75   |

-----  
Test on the values of the Statistic C

|                              |   |        |
|------------------------------|---|--------|
| Number of degrees of freedom | : | 89     |
| ChiSquare statistic          | : | 109.20 |
| p-value of test              | : | 0.07   |

-----  
CPU time used : 00:00:00.03

Generator state:

383498976 bits have been read.

===== Summary results of Rabbit =====

|                       |                     |
|-----------------------|---------------------|
| Version:              | TestU01 1.2.3       |
| Generator:            | ufile_CreateReadBin |
| Number of bits:       | 16777216            |
| Number of statistics: | 39                  |

Total CPU time: 00:00:05.97  
The following tests gave p-values outside [0.001, 0.9990]:  
(eps means a value < 1.0e-300):  
(eps1 means a value < 1.0e-15):

| Test                     | p-value  |
|--------------------------|----------|
| -----                    | -----    |
| 5 LinearComp             | 1 - eps1 |
| 22 MatrixRank, 320 x 320 | eps      |
| -----                    | -----    |

All other tests were passed